

Steam cooking significantly improves in vitro bile acid binding of collard greens, kale, mustard greens, broccoli, green bell pepper, and cabbage.



Bile acid binding capacity has been related to the cholesterol-lowering potential of foods and food fractions. Lowered recirculation of bile acids results in utilization of cholesterol to synthesize bile acid and reduced fat absorption. Secondary bile acids have been associated with increased risk of cancer. Bile acid binding potential has been related to lowering the risk of heart disease and that of cancer. Previously, we have reported bile acid binding by several uncooked vegetables. However, most vegetables are consumed after cooking. How cooking would influence in vitro bile acid binding of various vegetables was investigated using a mixture of bile acids secreted in human bile under physiological conditions. Eight replicate incubations were conducted for each treatment simulating gastric and intestinal digestion, which included a substrate only, a bile acid mixture only, and 6 with substrate and bile acid mixture. Cholestyramine (a cholesterol-lowering, bile acid binding drug) was the positive control treatment and cellulose was the negative control. Relative to cholestyramine, in vitro bile acid binding on dry matter basis was for the collard greens, kale, and mustard greens, 13%; broccoli, 10%; Brussels sprouts and spinach, 8%; green bell pepper, 7%; and cabbage, 5%. These results point to the significantly different ($P < \text{or} = .05$) health-promoting potential of collard greens = kale = mustard greens > broccoli > Brussels sprouts = spinach = green bell pepper > cabbage as indicated by their bile acid binding on dry matter basis. Steam cooking significantly improved the in vitro bile acid binding of collard greens, kale, mustard greens, broccoli, green bell pepper, and cabbage compared with previously observed bile acid binding values for these vegetables raw (uncooked). Inclusion of steam-cooked collard greens, kale, mustard greens, broccoli, green bell pepper, and cabbage in our daily diet as health-promoting vegetables should be emphasized. These green/leafy vegetables, when consumed regularly after steam cooking, would lower the risk of cardiovascular disease and cancer, advance human nutrition research, and improve public health.